Economics Task 6 — Balance of Payments & Terms of Trade

Short answer test: MCQ + short answer/data. Assesses five things — BOP concept and structure; reasons for Australia's CAB; the CAB and the savings/investment gap; the TOT and the TOT index; the effects of TOT changes.

1. Definitions — write these verbatim

BOP: a systematic record of all transactions between the residents of Australia and the residents of the rest of the world over a given period. *Residents* = individuals, firms, the government and organisations operating here.

Current account (CAB): the flow of goods, services and income between Australian residents and the rest of the world. "Current" because it's consumed/received in the current period.

KAFA: capital and financial transactions, including changes of ownership of Australia's assets and liabilities.

Terms of trade: an index measuring the relative movements in the prices of exports and imports — the ratio of export prices to import prices. It measures **the quantity of imports obtainable in exchange for a given volume of exports**.

S–I gap: the gap between Australia's intended investment and available domestic savings.

Servicing costs: future obligations from borrowing overseas (interest) or attracting foreign investment (dividends/profits).

2. Formulas

```
CREDIT = money INTO Australia      DEBIT = money OUT of Australia
BOGS (trade balance) = net goods + net services
CAB = BOGS + net primary income + net secondary income
CAB + KAFA = 0            ← always; double-entry book-keeping
CAB ≈ national savings (S) − investment (I)
TOT index   = (XPI ÷ MPI) × 100
Price index = (total this year ÷ total in base year) × 100
% change    = (new − old) ÷ old × 100        ← divide by the OLD value
```

Net errors and omissions is the balancing item, because some transactions are hard to measure or are missed.

3. Structure

Account	Component	Records	Examples
Current	**BOGS / trade balance**	Goods + services exported less imported	Iron ore, coal, LNG ↔ cars, fuel, machinery; tourism, education, freight
	Primary income	Income earned from, less paid to, the world from **working** and **investments**	Wages, dividends, interest, profits
	Secondary income	Government income/transfers where nothing is received in return	Foreign aid, pensions, insurance payouts, tax refunds
Capital	**Capital transfers**	Ownership transferred with nothing specific in return	Debt forgiveness, conditional grants for capital projects
	NPNF assets	**Intangibles** and rights to use natural resources	Brand names, copyrights, trademarks, mining/fishing rights
Financial	**Direct investment**	Long-term business investment, **≥10%** voting power	Buying a factory; 25% of a company
	Portfolio investment	Shares/bonds, **<10%**, no influence	5% of Facebook
	Financial derivatives	Contracts trading **risk**, not funds	Options on rates, currencies, indexes
	Reserve assets	RBA-held reserves	Gold, foreign currency
	Other investment	Residual	**Trade credit**, **currency and deposits**

4. Credits, debits and double entry

Current account: provide something of value → **credit**. Receive something of value → **debit**.

KAFA — counter-intuitive, learn it:

	Entry
Australia's liabilities **increase** OR assets **decrease**	**CREDIT (+)**
Australia's assets **increase** OR liabilities **decrease**	**DEBIT (−)**

Buy foreign shares → gain an asset → **debit**. Money leaves an Australian account to pay for imports → lose an asset → **credit**.

Double entry — the method: find the account for the **item**, then the offsetting **money** entry goes in the financial account with the **opposite sign**.

- Export → **CR** in CAB + **DR** in FA (currency)
- Import → **DR** in CAB + **CR** in FA (currency)

5. Classification drill — cover the right and work down

Transaction	Account	Sub-section	CR/DR
Coal export to China	Current	Goods	CR
Apple iPhone shipment to Australia	Current	Goods	DR
Mining company purchasing trucks	Current	Goods	DR
Chinese company paying for Australian education	Current	Services	CR
Australian spending money in Bali	Current	Services	DR
Purchasing shipping for iron ore	Current	Services	DR
Overseas company paying interest to an Australian company	Current	Primary income	CR
Australian backpacker earns money in London	Current	Primary income	CR
Businessman paying interest on a loan to an overseas investor	Current	Primary income	DR
US shareholder in Wesfarmers receives a dividend	Current	Primary income	DR
Donation to a charity in Africa	Current	Secondary income	DR
Chinese business buying 25% of an Australian company	Financial	Direct investment	CR
Australian company buying 5% of Facebook	Financial	Portfolio	DR
RBA selling gold overseas	Financial	Reserve assets	CR
Drawing on a UK bank account to buy an Australian house	Financial	Other investment	CR
Buying the rights to the Nando's name in Australia	**Capital**	NPNF assets	DR

Traps: capital *goods* (trucks, machinery) are still **goods** in the current account. Brand names/patents go in the **capital** account. 10% is the direct/portfolio cut-off.

6. BOP calculation — worked

Given: goods credits 600, goods debits 500, net services −300, net income ?, KAFA +1150.

```
CAB: CAB + KAFA = 0, so CAB = −1150         ← you need nothing else for this
Net income:  net goods = 600 − 500 = +100
             CAB = 100 + (−300) + net income = −200 + net income
             −200 + net income + 1150 = 0  →  net income = −950
```

Double-entry table — every transaction gives two entries and the table sums to zero. E.g. export $40,000 beef (CR goods, DR currency), $6,000 Fiji holiday (DR services, CR currency): CA = +$34,000, KAFA = −$34,000, total **$0**.

7. Reasons for Australia's CAB

CAB = BOGS + NPY + NSY. BOGS is volatile; **NPY is always in deficit**. **Often the NPY deficit exceeds the BOGS surplus → CAD.** Always address **both** components and say **which dominated** — that's where the marks are.

CAD = net debtor, investing more than saving. **CAS** = net creditor. A CAD is **not necessarily bad**.

Factor	Change	Chain	CAB
Trading partner growth (China)	↑	X ↑ (minerals, fuels, services) → credits ↑ → BOGS ↑	↑
Domestic growth	↑	income ↑ → consumption M ↑; output ↑ → capital/intermediate M ↑ → debits ↑ → BOGS ↓	↓
Exchange rate	AUD depreciates	X cheaper, M dearer → **if demand is price elastic** X ↑, M ↓ → BOGS ↑	↑
Terms of trade	↑	X prices ↑ relative to M prices → credits ↑ → BOGS ↑	↑
Competitiveness (relative wages, inflation differential, exchange rate)	↑	X more attractive → credits ↑ → BOGS ↑	↑
External shocks	—	COVID: X and M both fell, **M fell more** → BOGS ↑ → CAB ↑	either

Note: domestic growth moves the CAB the **opposite** way. State the **elasticity assumption** in the exchange rate chain.

NPY = servicing costs. Foreign **debt** → **interest**; foreign **equity** → **dividends and profits**. The deficit widens with a bigger S–I gap, a larger stock of foreign liabilities, higher interest rates, and higher profits of foreign-owned Australian firms.

Structural vs cyclical — use this framing, questions ask for it by name:

	STRUCTURAL	CYCLICAL
	Fundamental factors that mainly affect the **income balance**	Temporary factors, caused by the business cycle, that mainly affect the **trade balance**
	Savings–investment gap · foreign investment · foreign liabilities	Domestic business cycle · world business cycle · exchange rates · commodity prices · terms of trade

Structural factors are why the CAD **persists**; cyclical factors are why it **fluctuates**.

8. Savings–investment gap

> Australia has a **small population** and a **relatively low national savings rate** → a **gap between intended investment and domestic savings** → to close it Australia **borrows overseas (foreign debt)** and **sells Australian assets (foreign equity)** → both create **servicing obligations** (interest, dividends, profits) recorded as **primary income debits** → **NPY deficit widens → CAB falls**. The capital inflow funding the gap **is** the KAFA surplus — the mirror of the CAD.

Reverse: higher national savings → smaller gap → firms fund projects domestically → less foreign borrowing/equity → fewer NPY outflows → **CAB rises, CAD shrinks**, and the KAFA surplus shrinks.

Is a CAD a problem? Not if foreign capital funds **productive** investment returning more than its servicing cost, and if borrowing is voluntary private-sector activity (Pitchford "consenting adults"). It is if it funds **consumption**, or if rising servicing costs and reliance on capital inflow leave Australia exposed to a sudden stop or higher global rates. **It depends on what the capital is used for.**

9. Terms of trade — concept and index

Rise = FAVOURABLE: XPI up relative to MPI → buy **more** imports per unit of exports → purchasing power **up**. **Fall = UNFAVOURABLE:** the reverse. **The index level is meaningless — only the change matters.**

Building an index: total the prices each year; base year = 100; others = (total ÷ base total) × 100.

	Y1	Y2	Y3
Totals	$420	$410	$470
XPI	100	97.6	111.9

Worked calculation:

```
XPI 103, MPI 117 → (103/117) × 100 = 88.0
XPI 109, MPI 119 → (109/119) × 100 = 91.6
% change = (91.6 − 88.0) ÷ 88.0 × 100 = +4.1%  (favourable)
```

⚠ **Points vs per cent.** 95.5 → 97.4 is **1.9 index points** but **2.0 per cent** (1.9 ÷ 95.5). Read the wording; divide by the **old** value.

Four combinations — TOT rises when: XPI ↑ and MPI flat; MPI ↓ and XPI flat; **both ↑ but XPI ↑ more**; **both ↓ but MPI ↓ more**. (Flip for a fall.)

MCQ shortcut: `%ΔTOT ≈ %ΔXPI − %ΔMPI`. E.g. XPI +1.1%, MPI +5.7% → ≈ **−4.6%** (exact −4.35%).

10. Factors affecting the TOT

Australia is a **price taker** — X and M prices are set on world markets.

Commodity prices — the dominant driver. ~**75%** of exports are commodities. Demand **and** supply are **inelastic** → steep curves → **small shifts cause large price swings**. - *Inelastic demand:* few global substitutes; small share of final cost; necessity for production; long-term contracts. - *Inelastic supply:* long production lags; high capital requirements; geographic/climate constraints; regulatory and infrastructure limits. - Chain: `China/world growth ↑ → demand for commodities ↑ → prices ↑ → XPI ↑ → TOT ↑` (and reverse). Rising global supply → prices ↓ → XPI ↓ → TOT ↓. - *Diagram:* D1 → D2 rightward along a **steep S** → P1 → P2 and Q1 → Q2 → XPI ↑, CAB ↑, national income ↑.

Manufactured goods prices — most of our imports. Supply is **elastic** (flat curve) → prices stable and trending **down** (Chinese manufacturing, productivity, ICT technology) → MPI ↓ → **TOT ↑**. *Exception:* COVID supply bottlenecks → MPI ↑ → TOT ↓.

Oil prices — volatile, we're a major importer, and higher oil raises **energy and transport costs across all import prices**. But it also lifts our energy **export substitutes** (coal, LNG), so the net effect depends on weights.

Why the TOT tracks the XPI, not the MPI: the **supply of manufactured goods is more elastic than that of primary commodities**, so the XPI is far more volatile (range 60 vs MPI's 29).

11. Effects of a TOT rise

Same volume of exports earns more revenue. **Reverse all of these for a fall.**

Effect	Mechanism
BOGS ↑ → CAB ↑	Export credits rise relative to import debits
National income ↑, living standards ↑	Higher export revenue → higher incomes and profits; more imports per unit of exports → real purchasing power ↑
Investment ↑, unemployment ↓	Higher expected profitability of resource projects → investment ↑ → derived demand for labour ↑
Government revenue ↑	Company and income tax receipts and royalties ↑ → budget improves
AUD appreciates	Foreign buyers need more AUD; higher expected returns attract capital inflow → demand for AUD ↑
Inflation ↑	Higher income and profits → demand-pull pressure — *partly offset* because an appreciating AUD makes imports cheaper

Qualification worth a mark: the CAB improvement is **partly offset** because much of mining is **foreign-owned**, so higher profits mean higher **dividends paid overseas** → primary income debits ↑ → NPY deficit widens.

12. Trends over ten years

Period	Movement	Why
2014–16	TOT and CAB **falling**	Commodity prices collapse (iron ore US$180→US$40/t); China slowing; oil US$48→US$32; MPI stable
2016–19	TOT **+30%**, CAB **rising** but still CAD	XPI +40% (coal, iron ore, LNG); OPEC stabilised oil → LNG demand ↑. MPI rose by less. NPY deficit also rose (foreign-owned firms' profits)
2020	TOT small **fall**; CAB **rising**	COVID recession → XPI ↓ (LNG, coal); MPI ↓ (oil collapse, strong AUD). M fell more than X → BOGS ↑
2020–22	TOT **rising** to its peak of **115.6** (Jun qtr 2022)	Year to Sept 2021: **XPI +41% vs MPI +6.4%**. Recovery demand for raw materials; Ukraine war lifted oil → LNG/coal
Jun 2019 – Mar 2023	**Current account SURPLUS — first since 1975**	Record BOGS (**$30b** Jun 2021; peak **$40.6b** Jun 2022) on bulk commodity prices, iron ore driving a record **$53.3b** of metal ores exports. **NPY stayed in deficit** — the CAS was a BOGS story

Period	Movement	Why
2023–25	TOT and CAB **falling** → back to **CAD by 2024**	China slowdown → iron ore ↓ → XPI ↓ (MPI also fell, but XPI fell more). Recovery → income ↑ → M ↑. Foreign investment ↑ → dividends ↑ → NPY deficit ↑
2025–26	CAD **widening sharply**	Mar qtr 2026 CAD **$27.1b** (from $23.0b): primary income paid overseas **+$1.6b** on stronger dividends; trade surplus fell $5.2b to $3.9b; **first monthly goods deficit since Dec 2017**
Jun qtr 2026	TOT **−~4.4%**	**XPI +1.1% vs MPI +5.7%** (biggest MPI rise since Dec 2021 — fuel). Iron ore weak: Chinese stockpiles, strong global supply, anti-dumping

KAFA is the mirror of the CAB. Historically Australia has been a net recipient of capital inflow supplementing domestic savings → **KAFA surplus, the counterpart to our CAD**. During the 2019–23 CAS the KAFA moved into **deficit** (Australians investing more abroad than foreigners invested here). If a question tells you one, state the other.

13. Statistics to quote

CAD **$27.1b** (Mar qtr 2026), up from **$23.0b** · first monthly goods deficit since **Dec 2017** · **XPI +1.1% vs MPI +5.7%** (Jun qtr 2026) · TOT fell **≈4.4%** in Jun qtr 2026 · TOT trough **67.8** (Mar 2016) → peak **115.6** (Jun 2022) = **+70.5%**, then **−16.6%** to **96.4** (Mar 2026) · **XPI +41% vs MPI +6.4%** (year to Sept 2021) · CAS **Jun 2019 – Mar 2023**, first since **1975**, peak **+3.3% of GDP**; CAD now **−3.7% of GDP** · BOGS peak **$40.6b** (Jun 2022) · iron ore **US$180 → US$40/t** (2011–15) · commodities **~75%** of exports · XPI range **60** vs MPI range **29**

14. Exam technique

Command words: *Identify/State* — one word or figure. *Define* — formal wording. *Calculate* — formula, substitution, answer **with units**. *Describe* — what happened (direction, size, dates). *Distinguish* — both, with "whereas". *Explain* — why, as a **causal chain**. *Analyse* — components and relationships, using data. *Evaluate* — both sides plus a judgement.

Every "explain" answer: `cause → effect on X/M or credits/debits → effect on BOGS or NPY → effect on the CAB`

Calculations: write the formula first (worth a mark on its own), substitute visibly, give **units**, divide by the **old** value, check points vs per cent, state direction and **favourable/unfavourable**.

Traps. Points ≠ per cent. Price indexes tell you **nothing about values or volumes** — a favourite distractor. KAFA credit ≠ "money in". Capital goods are goods. ≥10% = direct investment. A falling TOT **reduces** growth and welfare, it doesn't raise them. State the elasticity assumption. A CAD isn't automatically bad.

Before handing in: units on every calculation · an arrow chain in every "explain" · a figure and a date in every trend answer · **both BOGS and NPY** in every CAB answer · favourable/unfavourable in every TOT answer.